

PYONGYANG SUNAN INTL, Korea, Democratic People's Republic ofWMO: **470580**Lat: **39.224N**Lon: **125.670E**Elev: **36**StdP: **100.90**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-12.7	-22.5	0.5	-11.6	-20.5	0.6	-9.8	8.5	-7.0	7.3	-6.9	1.7	320	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	31.7	25.0	30.5	24.2	29.4	23.7	26.5	30.0	25.7	28.9	25.1	27.9	2.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.7	28.7	24.8	19.9	27.7	24.2	19.2	27.2	83.1	30.2	79.8	29.1	77.0	28.0	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.0	5.8	4.9	DB	-17.2	33.6	2.9	1.2	-19.2	34.4	-20.9	35.1	-22.5	35.8	-24.6	36.7	(4)
(5)			WB	-17.7	27.5	2.6	1.2	-19.6	28.4	-21.1	29.1	-22.6	29.7	-24.5	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-5.4	-2.0	4.2	11.7	17.7	22.3	25.2	25.4	20.6	13.6	5.0	-2.6
	DBStd	11.29	4.16	4.05	3.62	3.64	2.70	2.21	2.02	2.17	2.63	3.69	4.75	4.37
	HDD10.0	1582	476	337	182	24	0	0	0	0	0	13	159	392
	HDD18.3	3199	734	570	437	201	44	2	0	0	10	152	399	650
	CDD10.0	2085	0	0	3	74	239	370	471	479	317	123	9	0
	CDD18.3	661	0	0	0	1	25	122	213	221	77	3	0	0
	CDH23.3	4726	0	0	0	10	194	734	1617	1801	357	13	0	0
	CDH26.7	1213	0	0	0	0	28	178	436	531	40	1	0	0
Wind	WSAvg	1.8	1.8	1.9	2.3	2.2	2.0	1.7	1.6	1.5	1.4	1.5	1.8	1.8
Precipitation	PrecAvg	1012	9	17	23	50	83	103	320	219	103	47	41	17
	PrecMax	1677	27	76	67	107	217	271	825	619	332	125	120	48
	PrecMin	585	0	0	0	0	12	26	90	29	16	1	4	2
	PrecStd	271	7	18	16	28	50	55	151	148	92	29	30	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	11.1	17.9	24.8	29.4	31.8	33.4	33.2	29.4	25.0	18.1	9.5
		MCWB	3.5	6.7	11.2	15.1	18.2	21.6	26.0	26.3	22.7	17.6	13.3	6.7
	2%	DB	4.0	8.0	14.7	22.3	27.1	30.1	31.5	31.9	27.8	22.6	15.5	7.0
		MCWB	1.7	4.2	8.6	13.5	17.6	21.0	25.0	26.0	21.5	16.8	11.9	4.9
	5%	DB	2.3	6.0	12.6	20.5	25.5	28.7	30.2	30.7	26.7	20.9	13.8	5.2
		MCWB	0.3	2.4	7.4	12.7	16.5	20.7	24.6	25.3	20.5	15.6	10.7	3.3
	10%	DB	0.8	4.1	10.5	18.6	23.9	27.3	29.1	29.6	25.5	19.3	12.0	3.5
		MCWB	-0.8	1.4	6.0	11.6	15.9	20.0	24.2	24.6	19.9	14.7	9.2	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	7.5	12.1	17.7	20.6	24.1	27.5	27.9	24.5	19.1	14.6	7.3
		MCDB	5.7	9.4	16.2	21.9	25.6	28.5	31.8	31.2	27.3	22.7	16.7	8.7
	2%	WB	2.0	4.7	9.8	15.2	19.1	22.9	26.3	26.8	23.2	17.6	12.6	5.2
		MCDB	3.7	7.7	13.5	20.6	24.6	27.3	29.6	30.3	25.9	21.0	15.0	6.8
	5%	WB	0.6	3.1	7.9	13.6	18.0	22.0	25.6	26.1	22.0	16.6	11.1	3.4
		MCDB	1.9	5.2	12.0	18.5	23.3	26.1	28.6	29.5	25.0	19.8	13.4	5.0
	10%	WB	-0.7	1.6	6.4	12.4	17.0	21.3	25.1	25.4	21.1	15.5	9.5	1.9
		MCDB	0.7	3.8	9.8	17.3	21.6	25.1	27.8	28.5	24.4	18.7	11.6	3.3
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.4	9.2	10.4	10.5	8.9	6.5	7.0	9.0	9.7	8.0	7.1
		MCDBR	8.9	10.7	12.5	13.4	13.2	11.5	8.4	8.2	9.9	10.9	9.9	8.5
		MCWBR	6.9	7.2	7.2	6.3	5.2	4.4	3.7	3.6	4.2	5.4	6.5	6.6
	5% WB	MCDBR	8.0	9.3	11.7	11.3	10.9	8.7	6.6	7.1	7.7	9.5	8.9	7.9
		MCWBR	6.7	7.0	7.6	6.3	5.2	4.3	3.5	3.6	4.1	5.5	6.7	6.6
Clear-Sky Solar Irradiance	taub	0.419	0.494	0.554	0.630	0.647	0.728	0.704	0.594	0.519	0.541	0.503	0.431	
	taud	1.930	1.796	1.708	1.612	1.625	1.536	1.626	1.848	1.954	1.802	1.833	1.931	
	Ebn at Noon	694	691	697	677	679	625	635	698	724	638	596	641	
	Edh at Noon	136	180	217	254	255	280	253	198	168	176	151	127	
All-Sky Solar Radiation	RadAvg	2.42	3.19	4.36	5.09	5.70	5.27	4.18	4.47	4.38	3.48	2.34	2.06	
	RadStd	0.17	0.23	0.33	0.40	0.51	0.54	0.55	0.52	0.37	0.22	0.24	0.14	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	+46

Nomenclature: See separate page